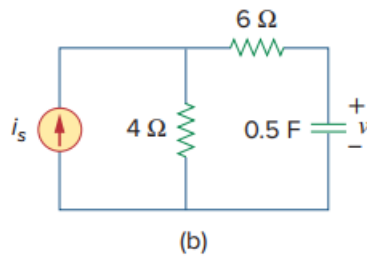
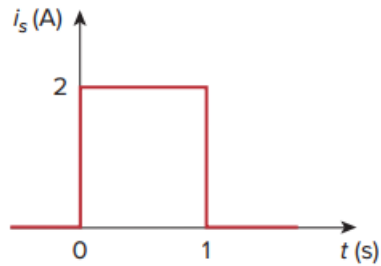


Problem

If the waveform in Fig. 7.116(a) is applied to the circuit of Fig. 7.116(b), find $v(t)$. Assume $v(0) = 0$.

Fig. 7.116:



Step-by-step solution

Step 1 of 7

Refer to Figure 7.116 in the textbook.

Consider for $t < 0$, the circuit becomes source free circuit and the voltage across capacitor is $v(0^-) = 0$.

Consider for $0 < t < 1$, i_s is applied to the circuit.

Determine the voltage at $t = 0^+$.

$$v(0^-) = 0$$

$$\begin{aligned} v(0^-) &= v(0^+) \\ &= 0 \text{ V} \end{aligned}$$

Step 2 of 7

Consider for $t = \infty$ the circuit reaches steady state.

Draw the circuit diagram in steady state as shown in Figure 1.

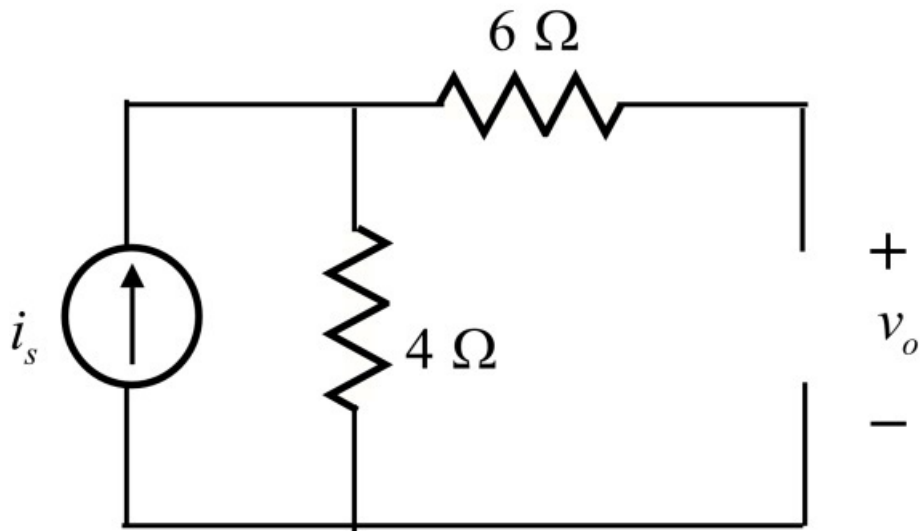


Figure 1

Step 3 of 7

Determine the voltage across capacitor at steady state.

$$\begin{aligned} v(\infty) &= i_s R \\ &= (2)(4) \\ &= 8\text{ V} \end{aligned}$$

Determine the equivalent resistance of the circuit at steady state.

At steady state $4\ \Omega$ and $6\ \Omega$ are in series.

$$\begin{aligned} R_{eq} &= 4 + 6 \\ &= 10\ \Omega \end{aligned}$$

Step 4 of 7

Determine the time constant of the circuit.

For RC circuit the time constant is,

$$\tau = R_{eq} C$$

Replace $10\ \Omega$ for R_{eq} , and 0.5 F for C in equation.

$$\begin{aligned} \tau &= (10)(0.5) \\ &= 5\text{ s} \end{aligned}$$

Consider the general equation for $v(t)$.

$$v(t) = v(\infty) + [v(0) - v(\infty)]e^{-\left(\frac{t}{\tau}\right)} \dots\dots (1)$$

Step 5 of 7

Determine the voltage $v(t)$ for $0 < t < 1$.

Replace 8 V for $v(\infty)$, 0 for $v(0)$, and 5 s for τ in equation 1.

$$\begin{aligned}v(t) &= 8 + [0 - 8]e^{-\left(\frac{t}{5}\right)} \\&= 8 - 8e^{-\left(\frac{t}{5}\right)} \\&= 8\left(1 - e^{-\left(\frac{t}{5}\right)}\right) \text{ V}\end{aligned}$$

Step 6 of 7

Consider for $t > 1$, the source current i_s is zero.

Determine the voltage $v(t)$ for $t > 1$.

$$v(t) = 8\left(1 - e^{-\left(\frac{t}{5}\right)}\right) \text{ V} \dots\dots (2)$$

Replace 1 for t in equation 2.

$$\begin{aligned}v(1) &= 8\left(1 - e^{-\left(\frac{1}{5}\right)}\right) \\&= 8(1 - e^{-0.2}) \\&= 1.45\end{aligned}$$

Determine the voltage at steady state for $t > 1$.

At steady state capacitor acts as open circuit.

$$v(\infty) = 0 \text{ V}$$

Step 7 of 7

Replace 0 V for $v(\infty)$, 1.45 for $v(1)$, and 5 s for τ in equation 1.

for $t > 1$, the initial condition is at $t = 1$ and t becomes $t - 1$.

$$\begin{aligned}v(t) &= 0 + [1.45 - 0]e^{-\left(\frac{t-1}{5}\right)} \\&= 1.45e^{-\left(\frac{t-1}{5}\right)}\end{aligned}$$

Therefore, the voltage across capacitor is

$$v(t) = \begin{cases} 8\left(1 - e^{-\left(\frac{t}{5}\right)}\right) \text{ V} & 0 < t < 1 \\ 1.45e^{-\left(\frac{t-1}{5}\right)} \text{ V} & t > 1 \end{cases}.$$